

CHE 201 Fall 2019 Quiz 2

Rija Firoz

TOTAL POINTS

85 / 100

QUESTION 1

1 85 pts

1.1 a 40 / 40

- ✓ - **0 pts** Correct
- **5 pts** h2 is not calculated or wrong
- **10 pts** KE is not calculated
- **5 pts** h1 is wrong
- **10 pts** PE not calculated

1.2 b+c 30 / 45

- **0 pts** Correct
- **45 pts** not done or wrong
- ✓ - **5 pts** unit conversion
- ✓ - **10 pts** b is wrong

QUESTION 2

2 2 15 / 15

- ✓ - **0 pts** Correct
- **15 pts** not done

NAME: Riya firoz UIN: _____

QUIZ 2: (100 pts) (Close book/closed notes) Show all your work for full credit

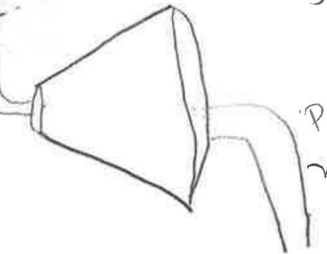
1. (85 points) The power output of steam turbine is 5MW, and the inlet and the exit conditions are as follow: $P_1=2\text{MPa}$, $T_1=400^\circ\text{C}$, $V_1=50\text{m/s}$, $z_1=10\text{m}$; $P_2=15\text{kPa}$, $x_2=0.90$, $V_2=180\text{m/s}$, $z_2=6\text{m}$. a) Compare the magnitudes of Δh , ΔKE , ΔPE . b) Determine the work done per unit mass of the steam flowing through the turbine. c) Calculate the mass flow rate of the steam.

2. (15 points) Briefly explain the process that Lanzatech patented. (Monday's guest speaker; Rachel Brenc)

$$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \sum_i \dot{m}_i \left(h_i + \frac{V_i^2}{2} + gz_i \right) - \sum_e \dot{m}_e \left(h_e + \frac{V_e^2}{2} + gz_e \right) \quad \dot{m} = \frac{AV}{v}$$

1 joule = 1 N.m

1 N = 1 kg.m/s²

$P_1 = 2\text{MPa} \times 10^6 = 2 \times 10^6 \text{ Pa}$
 $T_1 = 400^\circ\text{C}$
 $V_1 = 50\text{m/s}$
 $z_1 = 10$

 $P_2 = 15\text{kPa} = 15 \times 10^3 \text{ Pa}$
 $x_2 = 0.90$
 $V_2 = 180\text{m/s}$
 $z_2 = 6\text{m}$

5MW = $5 \times 10^6 \text{ W} = 5 \times 10^3 \text{ kW}$
 EM
 Steady State $dE_{cv}/dt = 0$
 One inlet / one outlet.
 $\dot{Q} = 0$
 a) Δh will increase because pressure decreases
 ΔKE = will increase because velocity decreases
 ΔPE = will increase because

$\frac{dE_{cv}}{dt} = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m} \left((h_1 - h_2) + \frac{(V_1^2 - V_2^2)}{2} + g(z_1 - z_2) \right)$
 because z value decreasing.

$0 = -5 \times 10^3 \text{ kW} + \dot{m} \left((h_1 - h_2) + \frac{(V_1^2 - V_2^2)}{2} + g(z_1 - z_2) \right)$
 $\dot{W} = 5\text{MW} = 5 \times 10^3 \text{ kW}$
 $= 5 \times 10^3 \frac{\text{kJ}}{\text{s}}$

$h_1 = 3247.6$

$h_2 = x h_g + (1-x) h_f$

$h_2 = 0.9 \times 1890.7 + (1-0.9) 908.79$

$h_2 = 1792.509$

$5 \times 10^3 = \dot{m} \left(3247.6 + 1792.509 \frac{\text{kJ}}{\text{kg}} + \frac{5^2 - 180^2}{2} \frac{\text{m}^2}{\text{s}^2} + 9.81(10 - 6) \frac{\text{kJ}}{10^3 \text{ J}} \right)$

$5 \times 10^3 \frac{\text{kJ}}{\text{s}} = \dot{m} (1455.091 + (-16187.5) + 39.24) \frac{\text{kJ}}{10^3 \text{ J}}$

$5 \times 10^3 = \dot{m} (-14.69)$

$\dot{m} = -340 \text{ kg/s}$

D) Lanza tech is a carbon recycling company and they patented the process of using carbon to make jet fuel. and rather than giving out carbon the process gives out water.